

## Group Project

**Details of the group project shall be presented and discussed separately.**

**Initial things to do (for now):**

1. Identify members of your group (**3-4 *persons per group***)
2. Install and learn R programming (or any modelling and simulation software)
3. Identify problem areas where MoSi may be applied.

# PROJECT OUTLINE

## 1. Project Overview

- **Problem Statement:** Brief overview of the real-world system being analyzed and why it requires intervention.
- **Project Goal:** The high-level objective (e.g., reduce queue times by 20%, optimize manufacturing throughput, simulate epidemic spread).
- **System Boundary:** A high-level description of what is included and excluded from the study.

## 2. Problem Formulation & Objectives

- **System Description:** Background details of the physical or conceptual system.
- **Modeling Objectives:** Specific questions the simulation is expected to answer.
- **Performance Measures (Outputs):** Quantitative metrics to evaluate success (e.g., wait time, resource utilization, failure rate).
- **Decision Variables (Inputs):**
- Factors you will intentionally change during experiments to test different strategies.

# 3. Conceptual Model Design

- **System Components & Entities:** The actors moving through the system (e.g., customers, data packets, components) and static assets (e.g., servers, machinery).
- **State Variables & Events:** Key states (busy, idle, blocked) and discrete events that trigger a change in system state.
- **Assumptions & Simplifications:** Necessary constraints made due to uncertainties, missing data, or to prevent over-complicating the model.
- **Logic Diagrams / Flowcharts:** Visual representation of the process flow (e.g., Rule Base Diagrams).

# 4. Data Collection & Input Modeling

- **Data Sources:** Where and how system data will be acquired (e.g., historical logs, field observations, sensor data).
- **Statistical Analysis:**

  Methods for identifying and handling missing or anomalous data.

  Fitting sample data to probability distributions (e.g., Exponential, Normal, Erlang) to handle system randomness.

# 5. Model Translation & Implementation

- **Simulation Tool Selection:** Justification for your tool choice based on system needs.

  *Software options*

  *Coding languages:* General-purpose scripting like Python (using SimPy) or MATLAB.

- **Architecture & Code Structure:** How logic, objects, and components (of the model) are tied together in software.

# 6. Verification and Validation (V&V)

- **Verification (Building the Model Right):** Debugging techniques, unit tests, code reviews, or visual animations to ensure the program behaves exactly as designed.
- **Validation (Building the Right Model):**
- Comparing preliminary simulation outputs against known, real-world historical data.
- Subject matter expert (SME) reviews to confirm system realism.

# 7. Results Analysis & Conclusion

- **Output Data Analysis:** Statistical comparison of different scenarios using metrics like confidence intervals and hypothesis testing.
- **Trade-off & Optimization Analysis:** Identifying bottlenecks, cost-benefit thresholds, or diminishing returns.
- **Final Recommendations:** Concrete decisions or adjustments suggested to stakeholders based on the data.

*8. Sample Outputs*

**Screenshots of outputs** with labels

*9. References/Sources*

Use IEEE citation format and style.

*10. Program Listing*

- Provide here the **listing** of the code/program.

**Note:** The project shall be presented by **all** the group members.